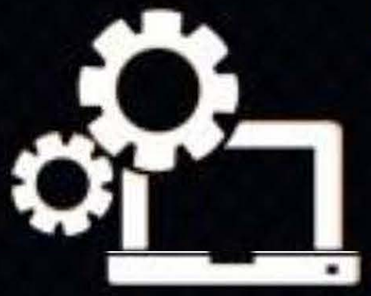
CS & IT ENGINEERING

DIGITAL LOGIC

Logic Gate



Lecture No. 4



By- CHANDAN SIR

Recap of Previous Lecture



NAND & NOR Gate

Symbol
Truth Table
Commutative & Associative Law Property

Topics to be Covered



Minimum Number of NAND & NOR Gate

Questions Practice

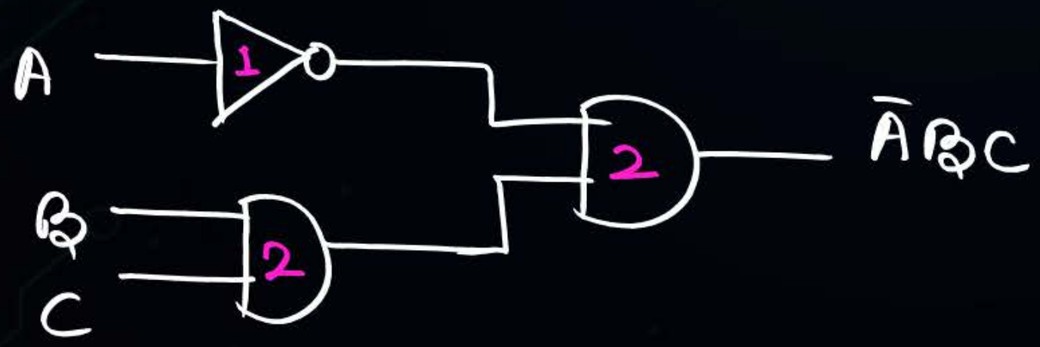
Case(1) $A \cdot B \cdot \bar{C} \cdot A \cdot \bar{E} \cdots$

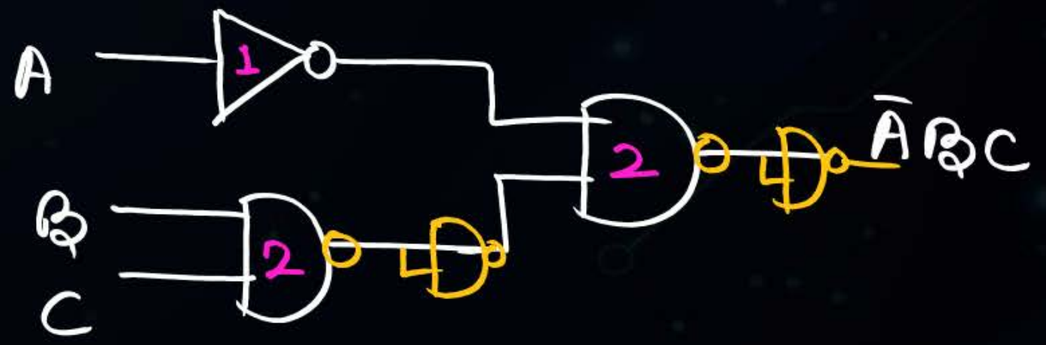
$n \rightarrow$ Total no. of Variables

$k \rightarrow$ Total no. of complement Variable

NAND	NOR
$(2n-2)+k$	$(3n-3)-k$

Ex $\bar{A}BC$ NAND = ?





1+2+2 = 5 Ans

NAND

$(2n-2)+k$

$n=3 \quad k=1$

$(2\times3-2)+1 = 4+1 = 5$



Ex $\bar{A}\cdot\bar{B}\cdot C$ NAND = ?



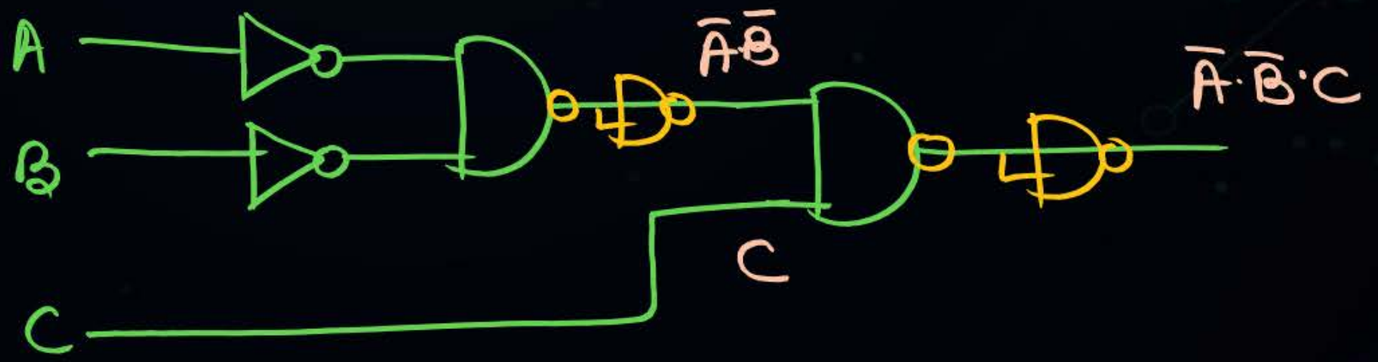
$n = 3$

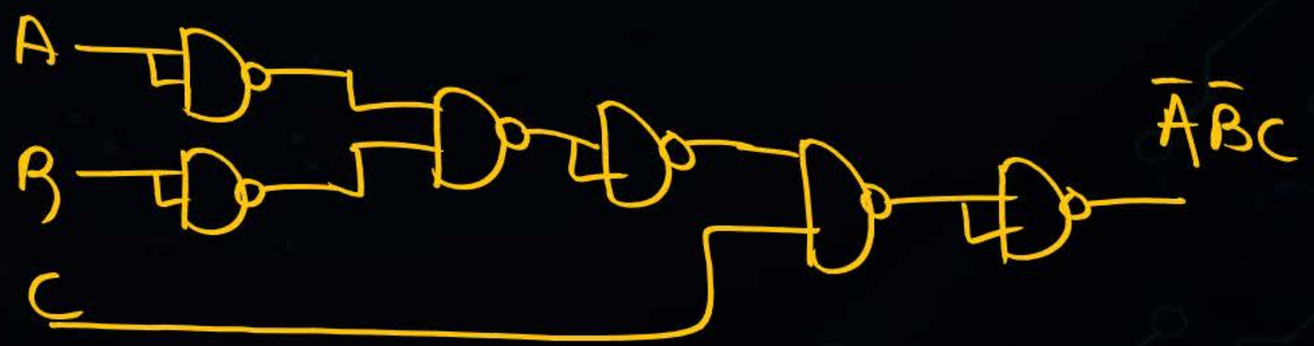
$K = 2$

$(2n-2)+K$

$(2\times3-2)+2$

$4+2 = 6$





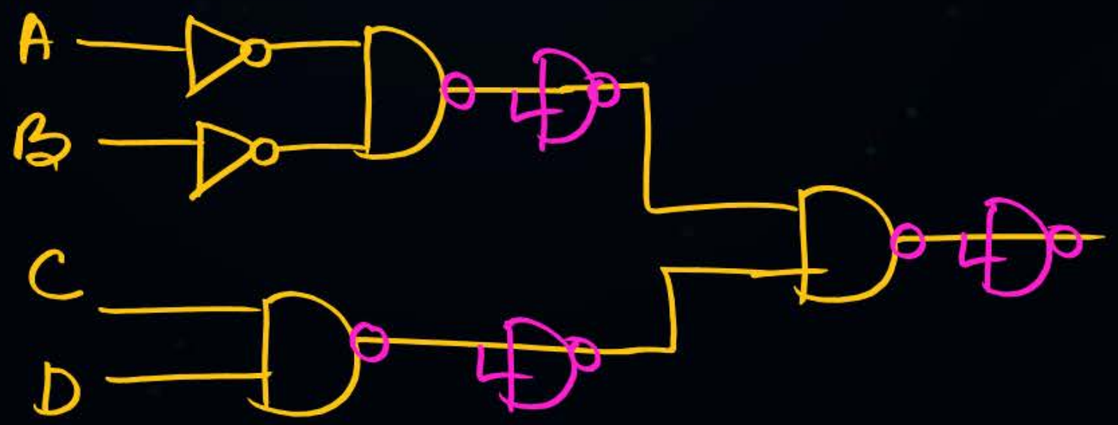
Q $\bar{A} \cdot \bar{B} \cdot C \cdot D$

$n = 4$

$k = 2$

$(2n-2)+k$

$6+2 = 8$ Ans



Q $\bar{A}\bar{B}\bar{C}\cdot D$

$n=4$

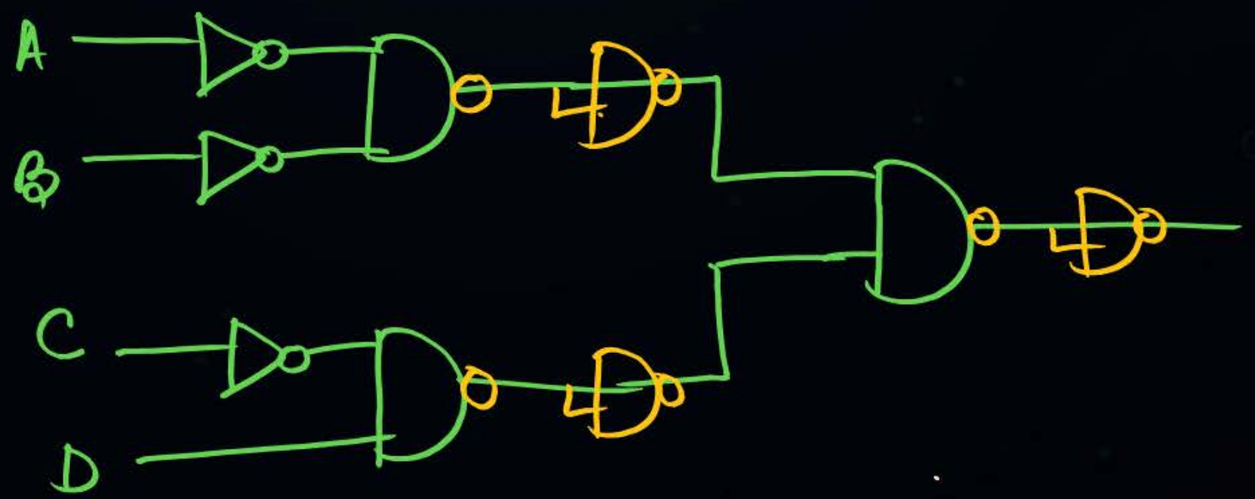
$k=3$

$(2n-2)+k$

$(2\times4-2)+3$

9

Ans



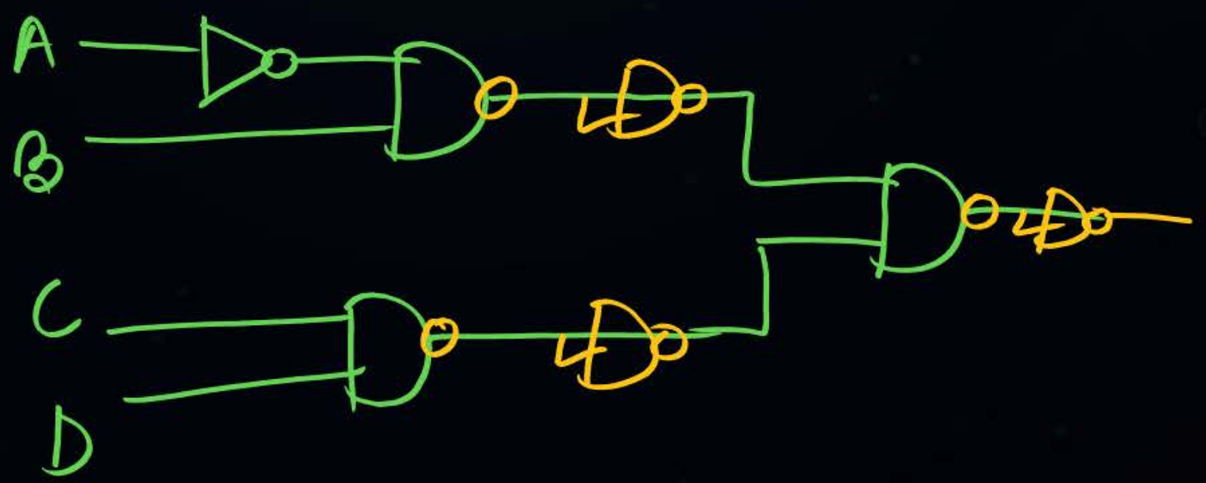
Q $\bar{A} \cdot B \cdot C \cdot D$

$n = 4$

$k = 1$

$(2n-2)+k$

7



Q $\bar{A} \cdot B \cdot C$ minimum no. Two input NOR GATE ?

$n = 3$

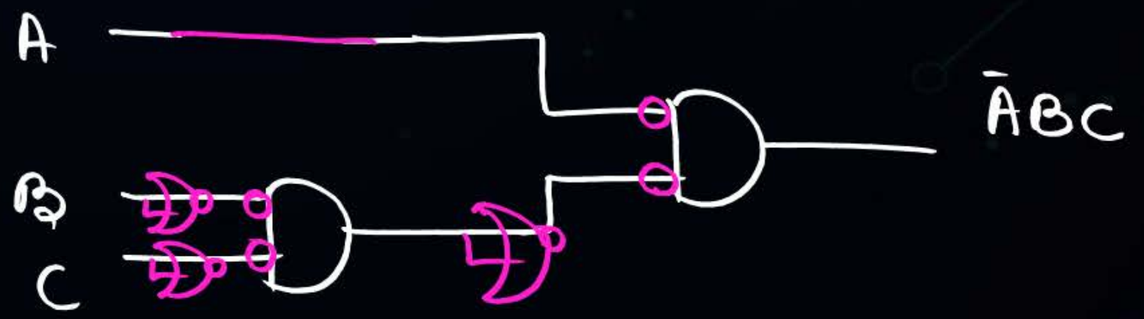
$K = 1$

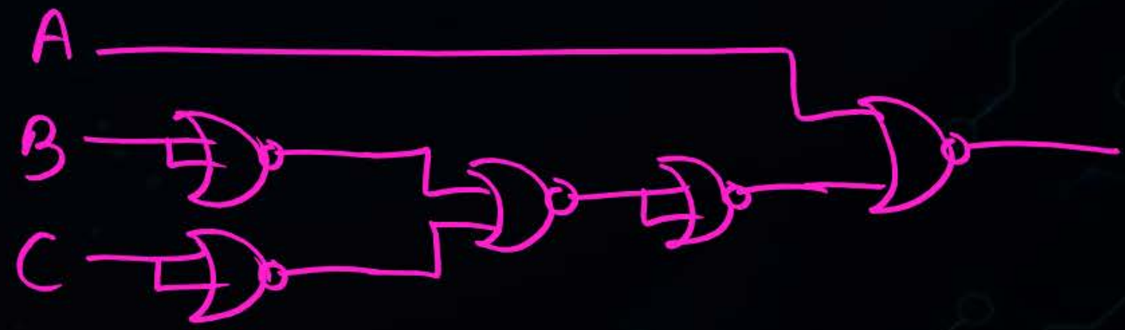
NOR $\Rightarrow (3n-3) - k$

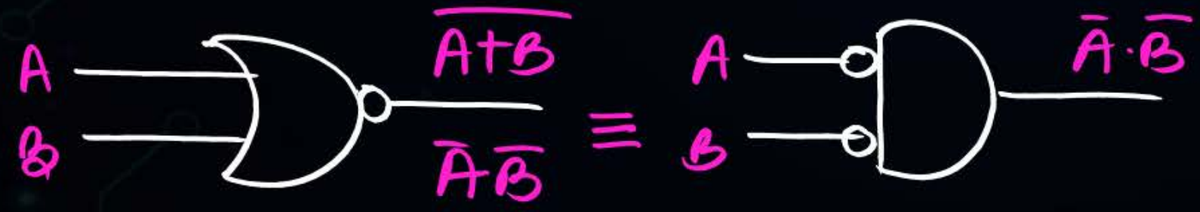
$= 9 - 3 - 1$

$= 5$








A
B
$\overline{A+B}$
$\overline{A}\,\overline{B}$
≡
A
B
$\bar{A}\cdot\bar{B}$


≡

Q $\bar{A} \cdot \bar{B} \cdot C$ minimum no. Two input NOR GATE ?

$n = 3$

$k = 2$

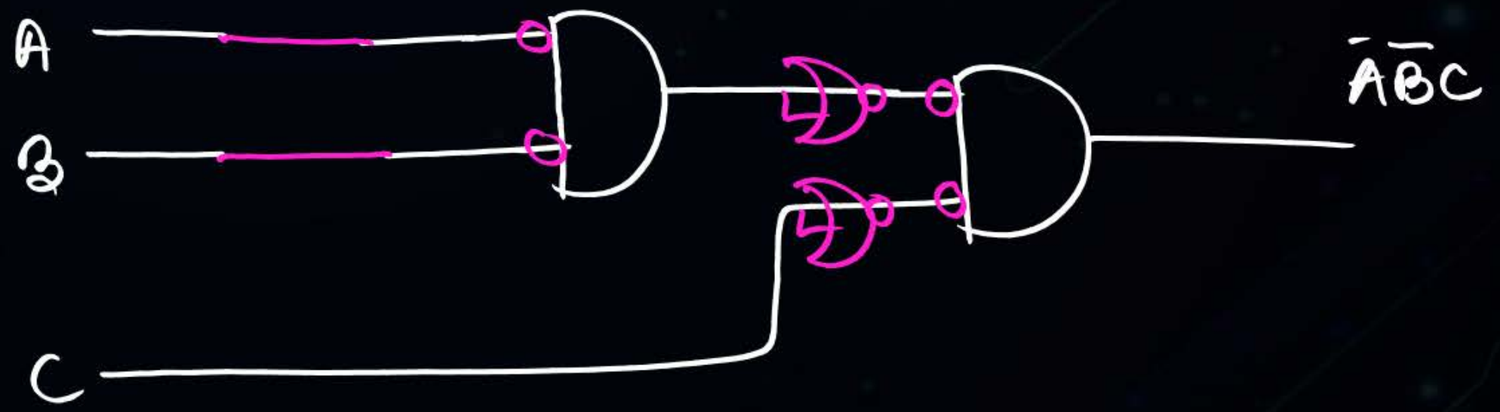
NOR $\Rightarrow (3n-3)-k$

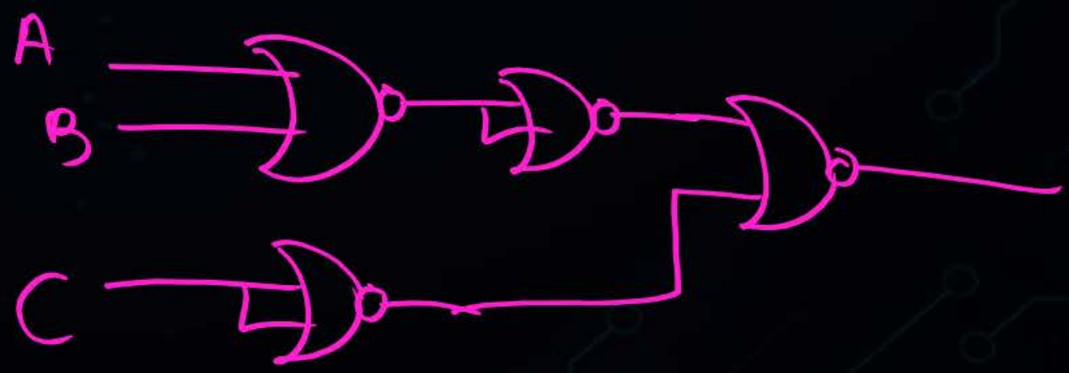
$= (3\times3-3)-2$

$= 6-2$

$= 4$

Ans





Q $\bar{A}\cdot\bar{B}\cdot C\cdot D$ minimum no. Two input NOR GATE ?

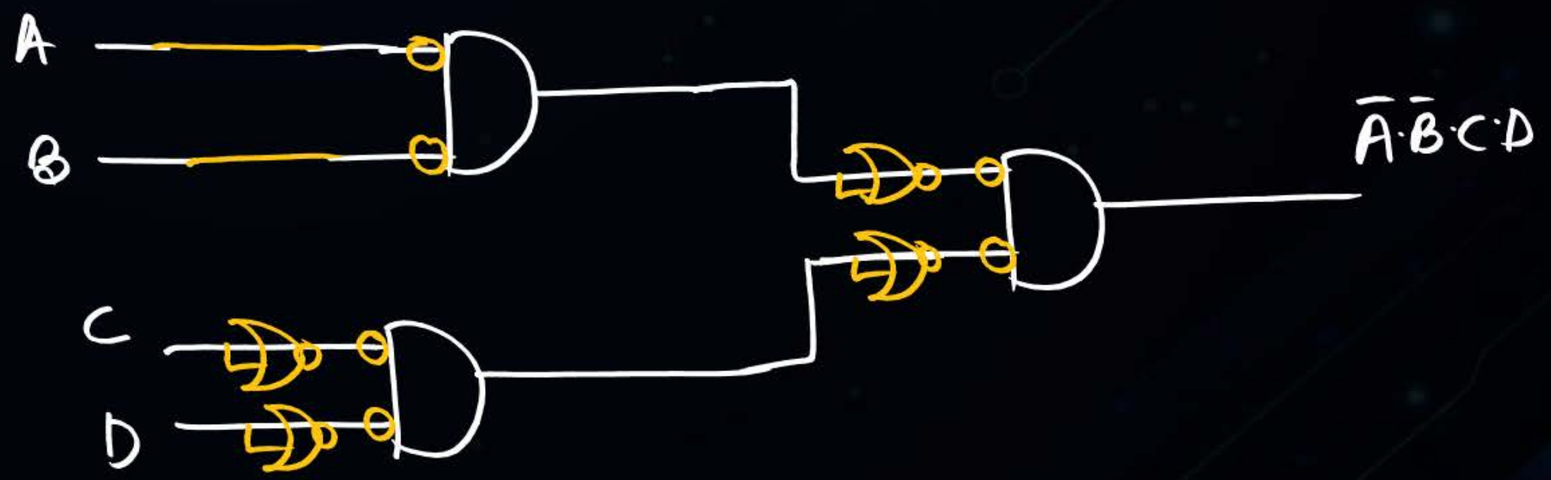
$n = 4$

$k = 2$

NOR $\Rightarrow (3n-3)-k$

$= (3\times4-3)-2$

$= 7$



Q $\bar{A}\bar{B}\bar{C}D$ minimum no. Two input NOR GATE ?

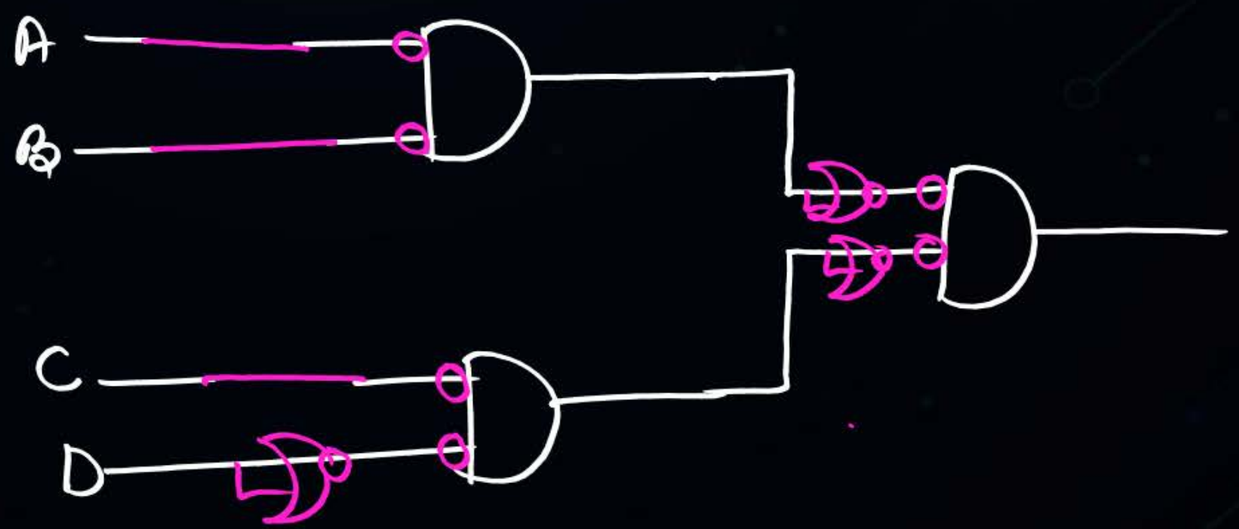
$n = 4$

$k = 3$

NOR $\Rightarrow (3n-3)-k$

$= (3 \times 4 - 3) - 3$

$= 6$



Q $\bar{A} \cdot B \cdot \bar{C} \cdot \bar{D} \cdot E \cdot F$

NAND = X

NOR = Y

Then $X + Y = ?$

(A) 25

(B) 23

(C) 20

(D) None

$n = 6 \quad k = 3$

$X = \text{NAND} = (2n-2)+k = 13$

$Y = \text{NOR} = (3n-3)-k = 12$

$X + Y = 13 + 12 = 25$

Ans

Case (2) $\bar{A}+B+\bar{C}+D+E+-+-+-+$

$n \rightarrow$ Total number of Variables

$K \rightarrow$ Total number of complement Variables

NAND	NOR
$(3n-3)-k$	$(2n-2)+k$

Q. $\bar{A}+B+C$ NAND = 2

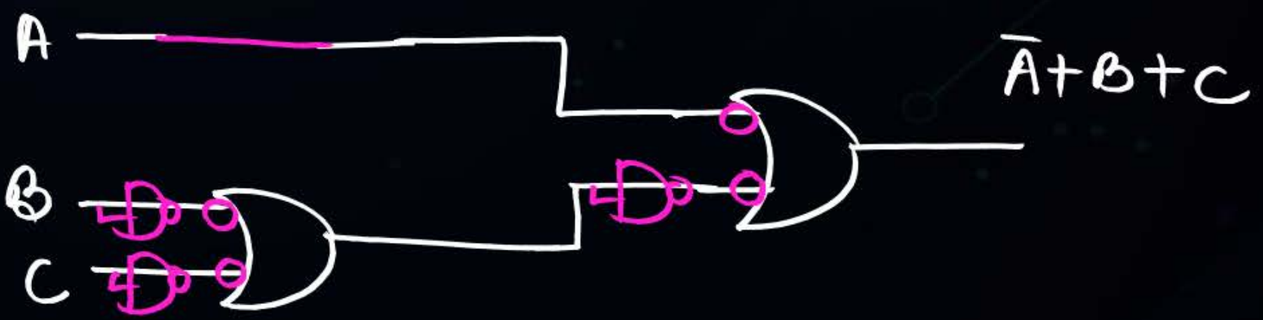
$n=3$

$k=1$

$NAND=(3n-3)-k$

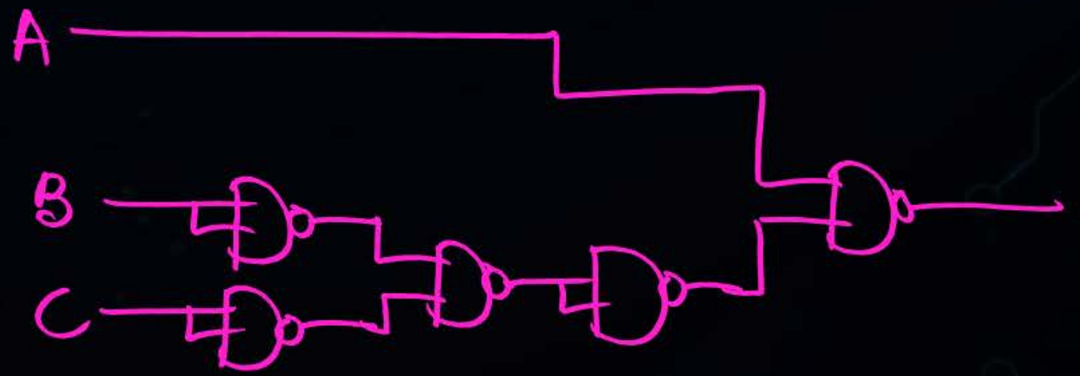
$=(3\times3-3)-1$

$=5$





NAND = Bubbled OR



Q $\bar{A}+B+C$

NOR = ?

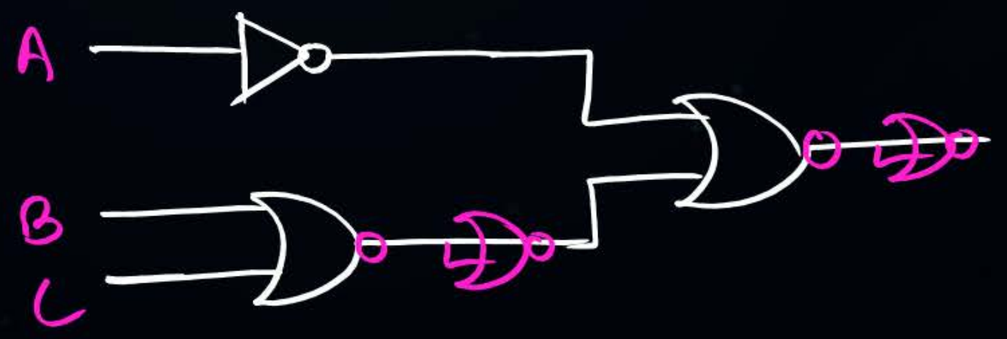
$n = 3$

$k = 1$

$NOR = (2n-2)+k$

$= 4+1$

$= 5$



$\bar{A}+B+C$

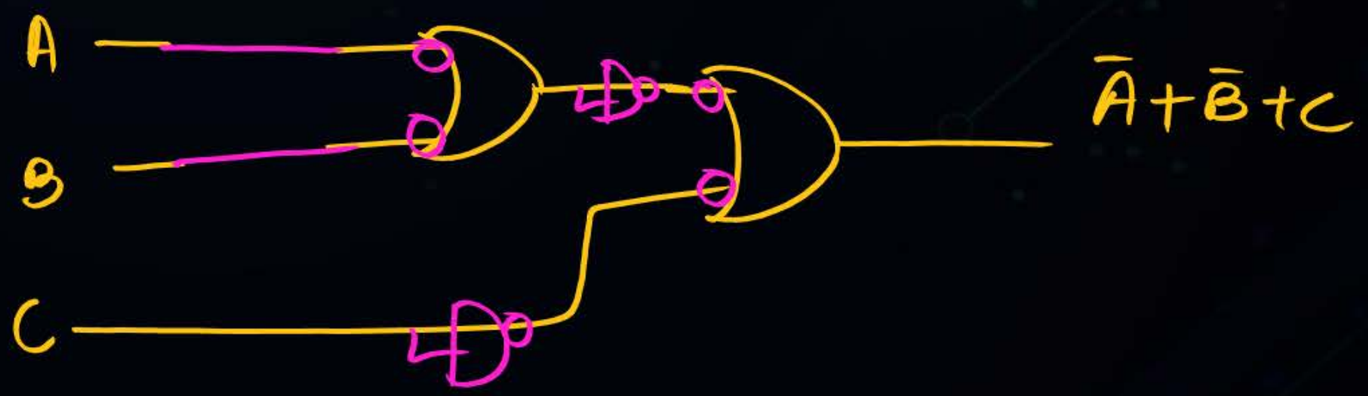
Q $\bar{A}+\bar{B}+C$

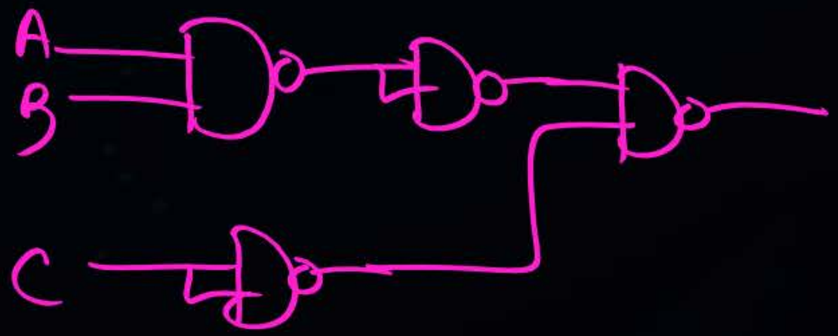
NAND = ?

n = 3 k = 2

NAND = (3n-3) - k

= 4





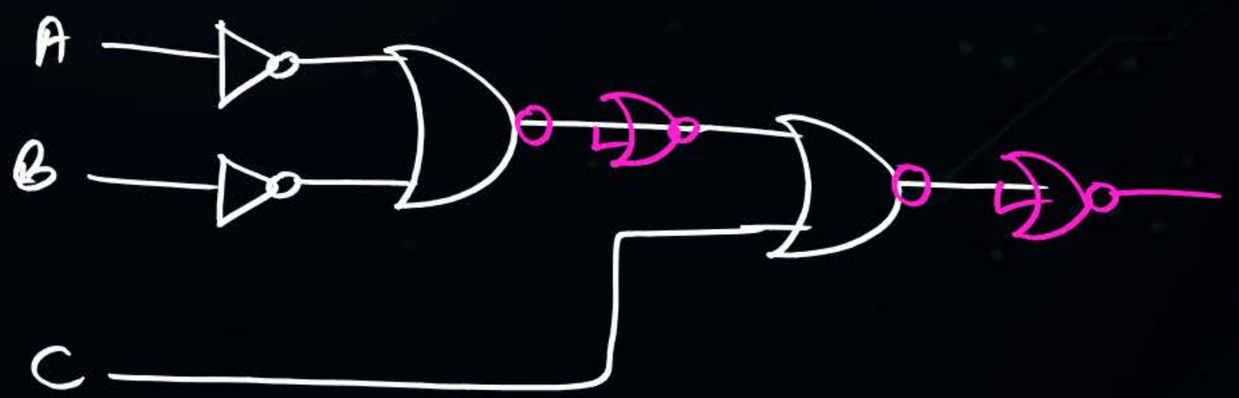
Q $\bar{A}+\bar{B}+C$

NOR = ?

NOR = (2n-2)+K

= (2X3-2) +2

= 6



#Q. The minimum number of two input NAND gates required to implement y = abcd is

$n=4 \quad K=0$

A 3

B 4

C 5

D 6

$NAND=(2n-2)+K$

$= 8-2=6$

#Q. Find the minimum number of two input NAND GATE required to implement the function-

$$f(A, B, C) = \bar{A}BC$$

A 3

B 6

C 5

D 4

$n = 3$

$k = 1$

$NAND = (2n-2) + k$

$= 4 + 1$

$= 5$

(MCQ)

#Q. Consider the following function f_1, f_2

$f_1 = \overline{A}\,\overline{B}\,\overline{C}\,D$

$f_2 = \overline{A}\,\overline{B}\,\overline{C}\,D + \overline{B}$

The number of NOR gate to implement f_1, f_2 will be n and m respectively. The value of n-m will be ?

$f_1 = \overline{A}\cdot\overline{B}\cdot\overline{C}\cdot D$

$n = 4 \quad k = 3$

$f_2 = \overline{A}\,\overline{B}\,\overline{C}\,D + \overline{B} \Rightarrow \overline{B}\,[\overline{A}\,\overline{C}\,D + 1] = \overline{B}$



$m = 1$

A 4

B 5

C 6

D 7

$NOR = (3n-3) + k$

$n = 6$

$n - m = 6 - 1 = 5$

Q $f = A + A\bar{B}C + ABC$

minimum number NAND GATE required to implement the function given below—

(A) 0

(B) 1

(C) 2

(D) Sir, Mai Tare jameen par hu.

$A + A\bar{B}C + ABC$

$A[1 + \underbrace{\bar{B}C + BC}_{\text{anything}}] = A \cdot 1 = A$



(NAT)

#Q. A logical function implementation is as given below.



The minimum number of 2-input NAND gates required to implement y is _____.

Q $\bar{A}+\bar{B}+C+D$

HW

Q $\bar{A} + \bar{B} + \bar{C} + D$

NAND = ?

NOR = ?



2 Minute Summary

Minimum Number of NAND & NOR Gate

Thank you DOSTO